

Manav Bhullar

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EDUCATION

Thapar Institute of Engineering and Technology

2023 – 2027

B.E. in Computer Engineering

- **Relevant Coursework:** Operating Systems, Object-Oriented Programming, Database Management Systems, Computer Networks, Software Engineering

ACHIEVEMENTS & CERTIFICATIONS

Solved **400+** **Data Structures & Algorithms** problems across LeetCode and GeeksforGeeks

Coursera: Advanced Relational Database and SQL, APIs in Node.js: RESTful Backend Development, CI/CD Pipeline with Docker

EXPERIENCE

Virsa Society, Thapar Institute

2023 – 2025

Technical Head (Core Member, 2023 – 2024)

- Approached 30+ local vendors in Patiala for stall partnerships at Virsa Mela, converting 20 into confirmed sponsors for an event drawing 4,000+ student attendees
- Rebuilt the society website (ssavirsa.in) frontend from scratch in **React**, migrating off a static HTML/CSS setup, and independently built the events and gallery pages

PROJECTS

SCALES v3.0 – Automated Short-Answer Grading Engine | *FastAPI, LiteLLM, HuggingFace Transformers, pytest*

- Designed a **Consistency-Based Trust Estimation (CBTE)** pipeline to eliminate a circular-dependency failure from v2.0, where confidence scoring had been trained on the LLM's own ~15,000 pseudo-labels and learned to trust its own hallucinations
- Built a **3-tier trust verification system**: exact evidence-span substring matching, **NLI entailment checking** via **cross-encoder/nli-deberta-v3-base**, and synonym-based stability re-prompting – deferring low-trust grades to human review instead of guessing
- Architected a model-agnostic grading pipeline via **LiteLLM** (Gemini 3.1 Flash-Lite), decomposing evaluation into independent **CERA** (concept extraction) → **CGR** (concept grading) → **CBTE** (trust scoring) → Aggregator stages
- Achieved deterministic, reproducible scoring with tuned trust thresholds ($\tau = 0.5$) and full **pytest** coverage across the grading pipeline

PIP-RAG – Placement Intelligence RAG System | *FastAPI, Qdrant, Gemini API*

- Built a conversational **RAG** system over TIET placement data, grounding LLM responses in retrieved interview questions and company data using **Qdrant** vector search and Gemini **text-embedding-004** embeddings
- Designed a custom API key rotation layer across 5–6 Gemini keys to scale free-tier throughput to ~7,500 requests/day, bypassing per-key rate limits
- Indexed 226 companies with metadata-grounded top-6 chunk retrieval and hard CGPA-eligibility filters applied directly at the vector search layer; exposed conversational Q&A, gap-analysis, and eligibility-shortlisting as separate FastAPI endpoints

Real-Time Ride-Matching Engine (Floq) | *Node.js, PostgreSQL, Redis, Socket.io* | Live

- Managed ride lifecycles utilizing a **Finite State Machine (FSM)**; prevented double-matching race conditions using Redis **SET NX PX** distributed locks (with Lua atomic release) and PostgreSQL **FOR UPDATE SKIP LOCKED** row-level locks
- Implemented a **read-through cache** pattern using Redis to optimize PostgreSQL read latency, and leveraged a **Pub/Sub** architecture via Socket.io for real-time event-driven dispatching; wrote **108 integration tests** with a 100% pass rate

TECHNOLOGIES

Languages: Python, JavaScript, C++

AI/ML: LiteLLM, HuggingFace Transformers, Gemini API, Qdrant (Vector DB), RAG, NLI/Embeddings, GitHub Copilot

Web & Backend: FastAPI, Node.js, Express.js, React.js, Socket.io

Database: PostgreSQL, MongoDB, Redis, Prisma ORM

Tools & Testing: Git, GitHub, Docker, pytest, Jest, Postman